ALGEBRA 1 · PERSONALIZED PRACTICE

Math Practice

2026-11-01 · 20 questions

I CAN USE INPUT-OUTPUT RULES TO REPRESENT RELATIONSHIPS.

1. I can use input-output rules to represent relationships.

A function machine triples the input and subtracts 2. What output comes from input 5?

2. I can use input-output rules to represent relationships.

Rules A and B are $2x + 1$ and $x + 3$. At which input do they give the same output?

3. I can use input-output rules to represent relationships.

A machine adds 4 and then doubles the result. The output is 18. Find the input.

4. I can use input-output rules to represent relationships.

A machine adds 4 to an input, then squares the result. Explain why $(x + 4)^2$ matches but $x^2 + 4$ does not.

I CAN DISTINGUISH FUNCTIONS FROM RELATIONSHIPS THAT ARE NOT FUNCTIONS.

5. I can distinguish functions from relationships that are not functions.

For $y = 2x - 5$, find the output when $x = -1$.

6. I can distinguish functions from relationships that are not functions.

Function or not?

$(0, 1), (2, 3), (0, 4)$

7. I can distinguish functions from relationships that are not functions.

Does $\{(1, 2), (1, 3), (2, 4)\}$ represent a function? Explain.

8. I can distinguish functions from relationships that are not functions.

Is $(1, 3), (2, 3), (3, 3)$ a function? Evaluate $2(-3) + 5$.

I CAN IDENTIFY THE DOMAIN AND RANGE OF A RELATION.

9. I can identify the domain and range of a relation.

x	-2	0	2	4
y	5	1	5	9

List the domain and range without repetitions.

10. I can identify the domain and range of a relation.

List the possible inputs.

A parking garage has levels 1 through 5. List the possible level inputs.

11. I can identify the domain and range of a relation.

List the domain.

$(2, 5), (4, 7), (6, 5)$

12. I can identify the domain and range of a relation.

For $f(x) = x + 2$ with inputs $\{-1, 0, 3\}$, list its outputs. Then extend 4, 9, 14, … one term.

I CAN CONNECT LINEAR GROWTH IN TABLES, GRAPHS, AND EQUATIONS.

13. I can connect linear growth in tables, graphs, and equations.

The outputs 2, 5, 8, 11 increase by a constant amount. What is the change per step?

14. I can connect linear growth in tables, graphs, and equations.

Find the change per step for outputs 5, 9, 13, 17. Is $(1, 2), (2, 2), (1, 3)$ a function?

15. I can connect linear growth in tables, graphs, and equations.

x	0	1	2	3
A	1	4	7	10
B	1	4	9	16

Which is linear? Give numerical evidence.

16. I can connect linear growth in tables, graphs, and equations.

Write an equation.

Slope 4; y-intercept −2.

I CAN CALCULATE AND INTERPRET SLOPE AS A CONSTANT RATE OF CHANGE.

17. I can calculate and interpret slope as a constant rate of change.

Find the slope.

(−2,1), (2,9)

18. I can calculate and interpret slope as a constant rate of change.

Find the slope through (0,3) and (4,11). Then evaluate $f(2)$ for $f(x) = 3x - 1$.

19. I can calculate and interpret slope as a constant rate of change.

One line has slope -2 and another has slope 3. Compare their directions and rates.

20. I can calculate and interpret slope as a constant rate of change.

As x rises by 4, y falls by 12. Find the slope.